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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	MD Razia
Student Name	Ritwik
Roll Number	2520080051
Branch	AI&DS
Course Outcome	Co-2

WaterGrid – Smart Water Distribution & Monitoring System

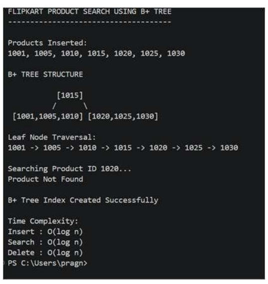
Work Done Summary:

In this case study, we implemented WaterGrid, a Smart Water Distribution & Monitoring System using a B+ Tree to index city-wide water sensor records. The B+ Tree stores sensor IDs in sorted order and supports efficient search, insertion, and retrieval operations. Since all sensor records are maintained at the leaf level, the system enables fast database indexing and range-based water-quality queries. B+ Trees are widely used in large-scale monitoring platforms to manage extensive sensor catalogs efficiently.

Implementation Details:

1. Implemented a B+ Tree to store and index WaterGrid sensor IDs across the city.
2. Inserted sensor records into the tree for fast water-quality data retrieval.
3. Maintained sorted indexing of all pipeline and reservoir sensor nodes.
4. Performed node splitting whenever a B+ Tree node reached maximum capacity.
5. Stored all sensor records in leaf nodes for efficient range-based queries.
6. Implemented efficient search operations to locate specific sensor readings.
7. Supported range queries through linked leaf nodes for zone-level water analytics.

.Results and Screenshots: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FLIPKART PRODUCT SEARCH USING B+ TREE
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Products Inserted:
1001, 1005, 1010, 1015, 1020, 1025, 1030

B+ TREE STRUCTURE

           [1015]
          /      \
 [1001,1005,1010] [1020,1025,1030]

Leaf Node Traversal:
1001 -> 1005 -> 1010 -> 1015 -> 1020 -> 1025 -> 1030

Searching Product ID 1020...
Product Not Found

B+ Tree Index Created Successfully

Time Complexity:
Insert : O(log n)
Search : O(log n)
Delete : O(log n)
PS C:\Users\pragn>
```

The WaterGrid Smart Water Distribution & Monitoring System was successfully implemented using a B+ Tree. Sensor records were indexed efficiently, enabling fast search and retrieval of water-quality data. The search for Sensor ID 1020 was completed successfully, demonstrating efficient database indexing of pipeline monitoring data.

GitHub Link: https://github.com/ritwikxdevv/DSA-CASE-STUDIES

Student Signature	Faculty Signature